

Fraction Model for the Numbers

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To the Instructor

These notes are intended for a course in which students may be proving theorems on their own for the first time. One axiomatic development for the numbers takes “*number*”, “ $<$ ”, “ $+$ ”, and “ $*$ ” as primitive words, and establishes axioms that guarantee a field structure for “ $+$ ” and “ $*$ ” on the set of numbers; establishes “ $<$ ” to be an order on the set of numbers which is Dedekind complete, has no maximum nor minimum, and admits a denumerable subset of the numbers which is dense in it; and connects the algebra of “ $+$ ” and “ $*$ ” to the geometry of “ $<$ ”.

For students who are truly naïve mathematically, this gives little context for making examples, since they may have only a vague recognition that the numbers they use as tools even have these properties. In these notes, the mathematical content addresses the question “Can we define objects, establish a comparison principle among them, and impose an algebra on them such that if the objects are interpreted as numbers, the comparison principle is used to define “ $<$ ”, and the algebra is used to define “ $+$ ” and “ $*$ ”, then the statements for the axioms can be demonstrated to be true for these meanings?”

In these notes, a fraction model is pursued. The students are given that the natural numbers exist, and questions of number theory may be deflected or pursued as the instructor chooses. Students are given the natural numbers, and addition, multiplication, subtraction, and division on the natural numbers, along with the admonition that whereas addition and multiplication always make sense, $5 - 3$ is okay but $3 - 5$ is not, and $9 \div 3$ makes sense, but $5 \div 3$ does not, since neither $3 - 5$ nor $5 \div 3$ is a natural number. My general choice is to accept those statements the students make about number theory that are actually true, as true.

I typically grant the infinitude of the primes, unique prime factorization, and the Euclidean algorithm. However, in a recent semester, a student proved the Euclidean algorithm as a part of an argument; I was glad I had not given the class that theorem. Sometimes I ask for a proof, particularly if a proof would be likely to afford an opportunity to teach a proof technique (such as finite induction). Working within the fraction model gives fruitful opportunities for application of number theory theorems, and students often make arguments about the number theory in the context of arguing about the model.

The problem set is designed so that even a class which plods will be exposed to ideas of comparing sets, imposing an order on a set, and imposing an algebra on a set. A class that experiences success from the beginning can be expected to get at least to the point of recognizing that the fraction model is not Dedekind complete. I teach these notes as a one-semester course. I have taught classes that extended the order to the completion, but I have never had a class get through the corresponding extension of the algebra to the completion.

The storyline for the course goes like this:

- Defining a fraction consists of specifying the numerator and denominator.
- With the comparison principle from classical geometry, the product of the extremes is less than the product of the means, different objects may not be comparable.
- Having corrected this flaw, the set admits an order with no minimum and no maximum, and with the property that between any two elements is another element.
- Using the algebra algorithms from grade school arithmetic and being careful, the set admits an order-preserving semigroup which lacks an identity element, and admits a multiplication which is a group and distributes over the semigroup.
- The set fails to have the Dedekind cut property for its comparison principle.
- Correct the flaw and make the necessary adjustments to the algebra.

Meanwhile, we are also counting sets.

- The natural numbers are shown to be infinite.
- Segments are shown to be as large as the entire set.
- The natural numbers are discovered to be as large as the (pre-Dedekind) set.
- The natural numbers are shown to be not as large as any Dedekind-complete set.

Instructors considering using these notes for a course should be aware of certain intentional idiosyncrasies built into them. I do not forbid my students to use books or any non-human sources they can find. I don't encourage it, either – if they ask if it is okay, I just tell them to make sure it doesn't keep them from solving problems. I do consciously make the problem set so that I think it is simpler to work on the problems directly than to find an appropriate source and make the necessary translations. Thus some very standard problems may be cast in molds that are not immediately recognizable. For instance, “the fraction with numerator 3 and denominator 4” is $(3, 4)$ in the model. If students don't realize this before encountering the density problem, sit back and enjoy how creative they get in building a replacement for the midpoint formula. The longer it takes, the more opportunities you will get to appreciate how rich the order structure for the rational numbers is when liberated from its algebra, because the students will think of the objects in ways those of us with too much education can't even imagine.

Of course, once a student makes this connection, grade-school arithmetic becomes an available source for intuition, and more rapid progress will be made through the notes. However, I consider having the student make that connection her/himself inviolate, and I would neither confirm nor deny the connection if asked. My standard response would be, “What I do know is that the definitions mean what the definitions say.” As another example, I use “ x and y are elements” and “each of x and y is an element” to distinguish contexts that are typically explained in books using the word *distinct*.

Besides being correct (*distinct* may also be used correctly, even if it is often used redundantly), the usage allows me to teach lessons about language I consider important. I use

the term “the numbers” where most sources use the term “the real numbers”. This is not important; I just do not like to miss a chance to voice my own personal prejudice that the complex numbers are every bit as real as the “real numbers”. I don’t want to be personally responsible for perpetuating the adjective “imaginary” when it might be interpreted by the unlearned as being an opposite in some sense to “real”.

Since progress through these notes depends on the students finding the conjectures which are not theorems, and then addressing the issues raised in trying to produce structures about which the conclusions *are* true, the instructor must exercise some care in *when* access to subsequent pages is granted to the students. Also, students often uncover theorems while working on problems, and theorems can often be sifted out of students’ arguments; these make nice addenda to the notes. I have included some examples from a past class of mine in Section 2. Keep in mind that these are merely statements of theorems that the students’ proofs turned up; the students may not have stated the theorem that way and the theorem itself may not have solved the problem it addressed.

Section 3 contains remarks about particular problems or definitions, and possible timing schemes for presenting the problems. I have taught Mathematics 315 at James Madison University, using some form of the notes to follow, many times. In this section, I have tried to give the benefit of some of my experience, without being too heavy-handed about what “ought” to happen. It is my hope that the remarks might help with the order in which the problems should be presented. The order I have given need not be the optimal sequence for a particular class.

I give only one test in this course: the final examination. I offer one credit each time a student presents an argument for a problem that the class judges as being correct. If a student has a problem that someone else presents, that student is allowed to turn in her/his write-up at the end of the class period in which the problem was finished. If the write-up is correct, the student gets one-half of a credit. To discourage note-taking and to encourage the students to think back through what they have seen, I offer myself as a source for any problem that has been solved by the class. This way, if a student cannot reproduce an argument, I am a safety net for their being able to see the proof again, and if necessary, again, and . . .

The final examination is given as a take-home, “use your own notes but nobody else’s”, test. It consists of a section of problems solved during the semester, and a section of problems that the students have either not yet solved or have not yet seen. Successful proofs for the problems proven during the course allow students to make a C or to keep whatever grade their work during the semester warranted. Successful work on the second section allows a student to increase her/his grade, or to atone for slip-ups on the first section.

To the Student

Introduction to students - An Introduction to Doing Mathematics

In this course, not only will you be responsible for understanding why the mathematics we cover is correct, but the responsibility for discovery will also be assigned to the class. One of the immediate results of this responsibility for doing mathematics yourself rather than just learning how someone else did it, will likely be an acute awareness of the difference between the challenge associated with understanding why something is correct, and discovering for yourself whether or not a conjecture is a theorem.

Doing mathematics can be extremely exhilarating when one succeeds in the discovery process; failing to do mathematics when one is putting in the time trying to do mathematics can be extremely frustrating. This introduction is intended to alert you to some tips that are designed to optimize the chances for success.

First, you must put in the time necessary to give your creative intelligence a chance to work. Flashes of insight typically occur after information is organized and mulled over. Commitment to solving problems often leads to help from the subconscious. Students often tell me that they got “the big idea” while walking across campus or after turning in for the night.

Second, solutions to problems need not come all at once. You may need to solve many small problems on the way to proving a theorem or disproving an incorrect conjecture. Some of the most important work in mathematics is the creation of technique. Take pride in progress toward a goal, as well as reaching the goal. Any information you uncover is more than you knew before, and solving a problem is usually just a matter of putting together enough small solutions to allow you to see why the big problem is correct.

Students often tell me that they would be glad to put in the time if they just knew where to start. The following scheme is offered toward that end.

The awareness stage:

1. Identify all the words in the problem, and make sure that you *know* the definition of each of them. Try to recall examples that have dealt with these notions before. If a definition is new, make some examples for the definition.
2. Identify any theorems that may have already dealt with ideas present in the problem.

Put techniques that gave rise to proofs in those contexts firmly in mind.

The direct approach:

3. Make an example that models the hypothesis to the problem, and try to show that the example exhibits the properties of the conclusion. (If you can prove that your example fails to have the properties of the conclusion, you will have shown that the problem is not a theorem!)
4. See if what allowed you to establish the conclusion in the example is a property of all examples covered by the hypothesis. If it is, write a proof. If not . . .
5. . . . make an example which models the hypothesis, but fails to have whatever special properties you used to get the conclusion in the previous example. Go to 3.

The indirect, or contrapositive approach:

6. Suppose that the conclusion is false, and try to show that the hypothesis must be false as well. If the problem is not a theorem, any conclusions you get must be qualities which an example that disproves the conjecture must have.
7. Try to be aware of properties that if they were added to the hypothesis, would guarantee the conclusion. Alternatively, you might also try to find conclusions that follow from the hypothesis, even if they do not include the one you seek. Even if you are not able to solve the problem as stated, you may be able to create a substitute theorem.

The main mindset is to be aware that even when arguments do not come quickly or easily, the hunt itself may be an important learning experience. Working on problems yourself is the central ingredient. Not only will it provide you with theorems that are “your own”, but even when someone beats you to a solution, it will put you in a much stronger position to analyze the argument given.

A Theory of Sets and Ordered Pairs

We will not create an axiomatic set theory. However, following is an idiomatic presentation of some conventions that axiomatic set theory implies. We presuppose the existence of formal English as a language for expressing properties.

The primitive words are “set”, “element”, “ordered pair”, “first coordinate”, and “second coordinate”.

- (i.) A set consists of an element or elements.
- (ii.) An element of a set, and the set consisting of that element, are different objects.
- (iii.) A set is defined by stating the properties its elements have. (The plural has been chosen here, but the definition of a set may be made by stating a single property, and a set may have a single element.)

- (iv.) Given a definition for a set, any object having the properties specified is an element of the set, and any element of the set has the properties specified in the definition.
- (v.) An ordered pair consists of a first coordinate and a second coordinate.
- (vi.) An ordered pair's first coordinate may be the same set-theoretic object as the second coordinate, but as a part of the ordered pair, being the first coordinate is distinguishable from being the second coordinate.

We reserve a notation for the creation of definitions of sets, and for defining ordered pairs.

Reserved symbols for definitions of sets are $\{ : \}$. A symbol is created to follow the open brace and precede the colon, and then properties that an element must have are stated in terms of that symbol after the colon and before the closed brace. Thus

$$\{x : x \text{ is a number and } x > 5\}$$

stands for “the set to which an element belongs provided that it is a number and it is greater than 5”.

Reserved symbols for definitions of ordered pairs are $(,)$. The first coordinate of the ordered pair is written after the open parenthesis and before the comma; the second coordinate of the ordered pair is written after the comma and before the closed parenthesis. Thus, $(p, 5)$ stands for the ordered pair whose first coordinate is p and whose second coordinate is 5.

The purpose of this course is to build a model for the numbers. Our ultimate goal is to prove that the statements which are typically taken as axioms for the numbers are theorems in our model. In an axiomatic treatment, “number”, “ $<$ ”, “ $+$ ”, and “ $*$ ” are taken as primitive words; thus we provide definitions within the model such that if they are interpreted as the primitive words, the statements made by replacing the primitive words in the axioms with their analogues in the model become the topics of consideration.

You may assume that the natural numbers exist, and have whatever properties number theory says they do. If there is doubt about a property of the natural numbers, we will either prove the property or indicate what property we are assuming.

Chapter 1

Problem Sequence

Definition 1: Suppose that each of X and Y is a set. The statement that f is a *function* from X into Y means that f is a set so that

- (i.) each element of f is an ordered pair whose first coordinate is an element of X and whose second coordinate is an element of Y ; and
- (ii.) if p is an element of X , then there is an element of f whose first coordinate is p ; and
- (iii.) if p and q are elements of f , then the first coordinate of p is not the first coordinate of q .

Notation: If f is a function from X into Y , and (p, q) is an element of f , then we may write $f(p) = q$.

Definition 2: Suppose that each of X and Y is a set, and that f is a function from X into Y . The statement that M is the *range* of f means that M is the set to which an element belongs provided that there is an element of f for which it is the second coordinate.

Problem 1: Suppose that X is a set with more than one element. (That X has more than one element means that if p is an element of X , then there is an element of X different from p .) Show that the set to which an element belongs provided that it is an ordered pair whose first coordinate is an element of X and whose second coordinate is an element of X , is not a function from X into X .

Definition 3: Suppose that X is a set, and that L is a set each element of which is an ordered pair whose first coordinate is an element of X and whose second coordinate is an element of X . The statement that L is an *order* on X means that

- (i.) if p is an element of X , then (p, p) is not an element of L ; and
- (ii.) if p and q are elements of X , then (p, q) is an element of L , or (q, p) is an element of L ; and
- (iii.) if (p, q) and (q, r) are elements of L , then (p, r) is an element of L .

Problem 2: Suppose that X is a set with exactly one element. Show that there is no order on X .

Problem 3: Suppose that X is a set with more than one element, and that L is an order on X . Show that L is not a function from X into X .

Definition 4: U is the set to which an element belongs provided that it is an ordered pair each of whose coordinates is a natural number.

Definition 5: Suppose that (a,b) and (x,y) are elements of U . The statement that (a,b) *precedes* (x,y) means that $a * y$ is less than $x * b$.

Definition 6: $G = \{(p,q) : p \text{ is an element of } U, \text{ and } q \text{ is an element of } U, \text{ and } p \text{ precedes } q\}$.

Problem 4: Suppose that x is an element of U . Show that there is an element of U , call such an element y , such that (x,y) is an element of G .

Problem 5: Suppose that x is an element of U . Show that there is an element of U , call such an element y , such that (y,x) is an element of G .

Problem 6: Suppose that x and y are elements of U , and (x,y) is an element of G . Show that there is an element of U , call such an element w , such that (x,w) and (w,y) are elements of G .

Problem 7: Show that G is an order on U .

Definition 7: Suppose that each of X and Y is a set. The statement that X *commands* Y means that there is a function from X into Y whose range is Y .

Problem 8: Show that U commands the natural numbers.

Definition 8: Suppose that each of X and Y is a set. The statement that X is a *subset* of Y means that if p is an element of X , then p is an element of Y .

Problem 9: Suppose that each of X and Y is a set, and that X is a subset of Y . Show that Y commands X .

Problem 10: Suppose that X and Y are sets, and that X is a subset of Y . Show that it is not the case that X commands Y .

Definition 9: Suppose that each of X and Y is a set, and that there is an element of X which is an element of Y . The *intersection* of X with Y is $\{x : x \text{ is an element of } X, \text{ and } x \text{ is an element of } Y\}$.

Notation: $X \cap Y$ stands for “the intersection of X with Y ”.

Definition 10: Suppose that X is a set, L is an order on X , and a and b are elements of X so that (a,b) is an element of L , and there is an element of X , call such an element c , so that (a,c) is an element of L , and (c,b) is an element of L . The *segment* from a to b by L is $\{x : (a,x) \text{ is an element of } L, \text{ and } (x,b) \text{ is an element of } L\}$.

Notation: If (a,b) is an element of the order L , then $\underline{a,b}$ stands for “the segment from a to b by L ”.

Problem 11: Suppose that x is an element of U . Show that there is a segment by G so that x is an element of it.

Problem 12: Suppose that X is a set, L is an order on X , (p, q) and (r, s) are segments by L , and x is an element of $(p, q) \cap (r, s)$. Show that $(p, q) \cap (r, s)$ is a segment by L .

Problem 13: Suppose that x and y are elements of U , and that G is an order on U . Show that there are segments by G , call them P and Q , so that

- (i.) x is an element of P ;
- (ii.) y is an element of Q ; and
- (iii.) if w is an element of P , then w is not an element of Q .

Definition 12: Suppose that each of X and Y is a set. The *union* of X with Y is $\{p : p \text{ is an element of } X \text{ or } p \text{ is an element of } Y\}$.

Notation: $X \cup Y$ stands for “the union of X with Y ”.

Definition 13: Suppose that X is a set, L is an order on X , and T and V are subsets of X . The statement that (T, V) is a *cut* of X by L means that

- (i.) $T \cup V = X$; and
- (ii.) if x is an element of T , and y is an element of V , then (x, y) is an element of L .

Problem 14: Suppose that X is a set, L is an order on X , and (A, B) is a cut of X by L . Show that if p is an element of A , then p is not an element of B .

Problem 15: Suppose that X is a set, and L is an order on X . Show that there is a cut of X by L .

Problem 16: Suppose that X is a set, L is an order on X , and (p, q) is an element of L . Show that there is a cut of X by L , call it (A, B) , so that p is an element of A and q is an element of B .

Definition 14: Suppose that X is a set, L is an order on X , p is an element of X , and M is a subset of X . The statement that p is the *max* of M by L means that p is an element of M , and if q is an element of M different than p , then (q, p) is an element of L . The statement that p is the *min* of M by L means that p is an element of M , and if q is an element of M different than p , then (p, q) is an element of L .

Definition 15: Suppose that X is a set, and L is an order on X . The statement that L has the *Dedekind cut property* means that if (A, B) is a cut of X by L , then

- (i.) A has a max by L , or B has a min by L ; and
- (ii.) it is not the case that both A has a max by L , and B has a min by L .

Problem 17: Suppose that $L = \{(x, y) : x \text{ is a natural number, } y \text{ is a natural number, and } x < y\}$. Show that L does not have the Dedekind cut property.

Definition 4': $U' = \{(x, y) : (x, y) \text{ is an element of } U; \text{ and if each of } a \text{ and } b \text{ is a natural number so that } a > 1 \text{ and } a * b = x, \text{ and each of } c \text{ and } d \text{ is a natural number such that } y = c * d, \text{ then } a \text{ is not } c \text{ and } a \text{ is not } d\}$.

Definition 6': $G' = \{(p, q) : p \text{ is an element of } U', q \text{ is an element of } U', \text{ and } p \text{ precedes } q\}$.

Problem 4': Suppose that x is an element of U' . Show that there is an element of U' , call such an element y , so that (x,y) is an element of G' .

Problem 5': Suppose that x is an element of U' . Show that there is an element of U' , call such an element y , so that (y,x) is an element of G' .

Problem 6': Suppose that x and y are elements of U' , and (x,y) is an element of G' . Show that there is an element of U' , call such an element w , so that (x,w) and (w,y) are elements of G' .

Problem 7': Show that G' is an order on U' .

Problem 8': Show that U' commands the natural numbers.

Problem 11': Suppose that G' is an order on U' , and that x is an element of U' . Show that there is a segment by G' such that x is an element of it.

Problem 18: Suppose that $T = \{(p,q) : (p,q) \text{ is an element of } U' \text{ and } p * p < 2 * q * q\}$ and $V = \{x : x \text{ is an element of } U' \text{ and } x \text{ is not an element of } T\}$. Show that (T,V) is a cut of U' by G' .

Problem 19: Suppose that $T = \{(p,q) : (p,q) \text{ is an element of } G' \text{ and } p * p < 2 * q * q\}$ and $V = \{x : x \text{ is an element of } G' \text{ and } x \text{ is not an element of } T\}$.

Possible conclusion i: T has a max by G' .

Possible conclusion ii: V has a min by G' .

Possible conclusion iii: T does not have a max by G' , and V does not have a min by G' .

Problem 20: Show that

Possible conclusion i: it is not the case that U' commands U .

Possible conclusion ii: U' commands U .

Problem 21: Show that

Possible conclusion i: it is not the case that $\mathbf{N}$ commands U' .

Possible conclusion ii: $\mathbf{N}$ commands U' .

Problem 22: Suppose that (x,y) is an element of G' . Show that

Possible conclusion i: it is not the case that $\underline{(x,y)}$ commands U' .

Possible conclusion ii: $\underline{(x,y)}$ commands U' .

Definition 16: $\rho = \{(((x,y), (p,q)), z) : (x,y) \text{ is an element of } U', (p,q) \text{ is an element of } U', \text{ and } z = ((x * q) + (p * y), y * q)\}$.

Problem 23: Show that ρ is a function from $\{(x,y) : x \text{ is an element of } U' \text{ and } y \text{ is an element of } U'\}$ into U' .

Definition 17: $\tau = \{(((x,y), (p,q)), z) : (x,y) \text{ is an element of } U', (p,q) \text{ is an element of } U', \text{ and } z = (x * p, y * q)\}$.

Problem 24: Show that τ is a function from $\{(x,y) : x \text{ is an element of } U' \text{ and } y \text{ is an element of } U'\}$ into U' .

Definition 18: Suppose that (x,y) is an element of U' . $EC_{(x,y)} = \{(p,q) : \text{there is a natural number, call it } k, \text{ such that } p = k * x \text{ and } q = k * y\}$.

Problem 25: Suppose that (x,y) is an element of U . Show that there is an element of U' , call it w , so that (x,y) is an element of EC_w .

Problem 26: Suppose that (p,q) and (x,y) are elements of U' and that (r,s) is an element of $EC_{(p,q)}$. Show that (r,s) is not an element of $EC_{(x,y)}$.

Problem 27: Suppose that (p,q) and (x,y) are elements of U' , so that $((p,q), (x,y))$ is an element of G' , and (r,s) is an element of $EC_{(p,q)}$ and (w,z) is an element of $EC_{(x,y)}$. Show that $((r,s), (w,z))$ is an element of G .

Definition 16': $\rho' = \{(((x,y), (p,q)), z) : (x,y) \text{ is an element of } U', \text{ and } (p,q) \text{ is an element of } U', \text{ and } z \text{ is an element of } U', \text{ and } ((x * q) + (p * y), y * q) \text{ is an element of } EC_z\}$.

Problem 23': Show that ρ' is a function from $\{(x,y) : x \text{ is an element of } U' \text{ and } y \text{ is an element of } U'\}$ into U' .

Definition 17': $\tau' = \{(((x,y), (p,q)), z) : (x,y) \text{ is an element of } U', \text{ and } (p,q) \text{ is an element of } U', \text{ and } z \text{ is an element of } U', \text{ and } (x * p, y * q) \text{ is an element of } EC_z\}$.

Problem 24': Show that τ' is a function from $\{(x,y) : x \text{ is an element of } U' \text{ and } y \text{ is an element of } U'\}$ into U' .

Problem 28: Suppose that x and y are elements of U' . Show that $\rho'((x,y)) = \rho'((y,x))$.

Problem 29: Suppose that x and y are elements of U' . Show that $\tau'((x,y)) = \tau'((y,x))$.

Problem 30: Suppose that each of x, y , and z is an element of U' . Show that $\rho'((\rho'((x,y)), z)) = \rho'((x, \rho'((y,z))))$.

Problem 31: Suppose that each of x, y , and z is an element of U' . Show that $\tau'((\tau'((x,y)), z)) = \tau'((x, \tau'((y,z))))$.

Problem 32: Suppose that each of x, y , and z is an element of U' . Show that $\tau'((x, \rho'((y,z)))) = \rho'((\tau'((x,y)), \tau'((x,z))))$.

Problem 33: Show that there is an element of U' , call such an element ζ , such that if x is an element of U' , then $\rho'((x, \zeta)) = x$ and $\rho'((\zeta, x)) = x$.

Problem 34: Show that there is an element of U' , call such an element μ , so that if x is an element of U' , then $\tau'((x, \mu)) = x$ and $\tau'((\mu, x)) = x$.

Problem 35: Suppose that x and y are elements of U' , (x,y) is an element of G' , and z is an element of U' . Show that $(\rho'((x,z)), \rho'((y,z)))$ is an element of G' .

Problem 36: Suppose that x and y are elements of U' so that (x,y) is an element of G' . Show that there is exactly one element of U' , call it q , such that $\rho'((x,q)) = y$.

Problem 37: Suppose that x is an element of U' . Show that there is exactly one element of U' , call it y , so that $\tau'((x,y)) = (1,1)$.

Definition 4'': $U'' = \{x : x \text{ is an element of } U'; \text{ or there is a cut of } U' \text{ by } G', \text{ call it } (A,B), \text{ so that } A \text{ has no maximum by } G' \text{ and } B \text{ has no minimum by } G', \text{ and } x = A\}$.

Definition 6'': $G'' = \{(x,y) : x \text{ and } y \text{ are elements of } U''; \text{ and if } x \text{ and } y \text{ are elements of } U', \text{ then } (x,y) \text{ is an element of } G', \text{ or if } x \text{ is an element of } U' \text{ and } y \text{ is not an element of } U', \text{ then } x \text{ is an element of } y, \text{ or if } x \text{ is not an element of } U' \text{ and } y \text{ is an element of } U', \text{ then } y \text{ is not an element of } x, \text{ or if neither } x \text{ nor } y \text{ is an element of } U', \text{ then } x \text{ is a subset of } y\}$.

Problem 4'': Suppose that x is an element of U'' . Show that there is an element of U'' , call such an element y , so that (x,y) is an element of G'' .

Problem 5'': Suppose that x is an element of U'' . Show that there is an element of U'' , call such an element y , so that (y,x) is an element of G'' .

Problem 6'': Suppose that x and y are elements of U'' , and (x,y) is an element of G'' . Show that there is an element of U' , call such an element w , so that (x,w) and (w,y) are elements of G'' .

Problem 7'': Show that G'' is an order on U'' .

Problem 38: Show that G'' has the Dedekind cut property.

Problem 22'': Suppose that (x,y) is an element of G'' . Show that $\underline{(x,y)}$ commands U'' .

Definition 19: Suppose that X is a set, L is an order on X , and (p,q) is an element of L . The *interval* from p to q by L is $\{x : x \text{ is an element of } \underline{(p,q)}, \text{ or } x \text{ is } p, \text{ or } x \text{ is } q\}$.

Notation: If (p,q) is an element of the order L , $[p,q]$ stands for “the interval from p to q by L ”.

Definition 20: Suppose that X is a set. The statement that s is a *sequence* in X means that s is a function from the natural numbers into X .

Problem 39: Suppose that $M = \{x : \text{there is an element of } G'', \text{ call it } (p,q), \text{ so that } x = [p,q]\}$. Show that there is a sequence in M , call such a sequence f , so that if n is a natural number, then $f(n+1)$ is a subset of $f(n)$.

Problem 40: Suppose that (A,B) is an element of G'' , and x is an element of U'' so that x is an element of (A,B) . Show that there is an element of G'' , call it (p,q) , so that $[p,q]$ is a subset of $\underline{(A,B)}$ and x is not an element of $[p,q]$.

Problem 41: Suppose that $M = \{x : \text{there is an element of } G'', \text{ call it } (p,q), \text{ so that } x = [p,q]\}$; s is a sequence in M so that if k is a natural number, then $s(k+1)$ is a subset of $s(k)$; and $A = \{x : \text{there is a natural number, call it } k, \text{ so that if } p \text{ is an element of } s(k), \text{ then } (x,p) \text{ is an element of } G''\}$. Show that $(A, \{x : x \text{ is an element of } U'' \text{ and } x \text{ is not an element of } A\})$ is a cut of U'' by G'' .

Problem 42: Suppose that $M = \{x : \text{there is an element of } G'', \text{ call it } (p, q), \text{ so that } x = [p, q]\}$; and s is a sequence in M so that if k is a natural number, then $s(k+1)$ is a subset of $s(k)$. Show that there is an element of U'' , call it w , so that if k is a natural number, then w is an element of $s(k)$.

Problem 43: Show that the natural numbers do not command U'' .

Problem 44: Suppose that (A, B) is a cut of U' by G' so that A has no max by G' , p is an element of U' , and $C = \{x : x \text{ is an element of } A, \text{ or there is an element of } A, \text{ call it } q, \text{ so that } x = \rho'((p, q))\}$. Show that C has no max by G' .

Problem 45: Suppose that (A, B) is a cut of U' by G' such that A has no max by G' , p is an element of U' , and $C = \{x : x \text{ is an element of } A, \text{ or there is an element of } A, \text{ call it } q, \text{ so that } x = \rho'((p, q))\}$. Show that $(C, \{x : x \text{ is an element of } U' \text{ and } x \text{ is not an element of } C\})$ is a cut of U' by G' .

Problem 46: Suppose that (A, B) is a cut of U' by G' so that A has no max by G' , that (D, E) is a cut of U' by G' so that D has no max by G' , and that $C = \{x : x \text{ is an element of } A, \text{ or there is an element of } A, \text{ call it } p, \text{ and an element of } D, \text{ call it } q, \text{ such that } x = \rho'((p, q))\}$. Show that C has no max by G' .

Problem 47: Suppose that (A, B) is a cut of U' by G' so that A has no max by G' , that (D, E) is a cut of U' by G' so that D has no max by G' , and that $C = \{x : x \text{ is an element of } A, \text{ or there is an element of } A, \text{ call it } p, \text{ and an element of } D, \text{ call it } q, \text{ such that } x = \rho'((p, q))\}$. Show that $(C, \{x : x \text{ is an element of } U' \text{ and } x \text{ is not an element of } C\})$ is a cut of U' by G' .

Problem 48: Suppose that (A, B) is a cut of U' by G' so that A has no max by G' , p is an element of U' , and $C = \{x : \text{there is an element of } A, \text{ call it } q, \text{ so that } x = \tau'((p, q))\}$. Show that C has no max by G' .

Problem 49: Suppose that (A, B) is a cut of U' by G' so that A has no max by G' , p is an element of U' , and $C = \{x : \text{there is an element of } A, \text{ call it } q, \text{ so that } x = \tau'((p, q))\}$. Show that $(C, \{x : x \text{ is an element of } U' \text{ and } x \text{ is not an element of } C\})$ is a cut of U' by G' .

Problem 50: Suppose that (A, B) is a cut of U' by G' so that A has no max by G' , that (D, E) is a cut of U' by G' so that D has no max by G' , and that $C = \{x : \text{there is an element of } A, \text{ call it } p, \text{ and an element of } D, \text{ call it } q, \text{ such that } x = \tau'((p, q))\}$. Show that C has no max by G' .

Problem 51: Suppose that (A, B) is a cut of U' by G' so that A has no max by G' , that (D, E) is a cut of U' by G' so that D has no max by G' , and that $C = \{x : \text{there is an element of } A, \text{ call it } p, \text{ and an element of } D, \text{ call it } q, \text{ so that } x = \tau'((p, q))\}$. Show that $(C, \{x : x \text{ is an element of } U' \text{ and } x \text{ is not an element of } C\})$ is a cut of U' by G' .

Definition 16'': $\rho'' = \{(x, y), z) : x \text{ is an element of } U'' \text{ and } y \text{ is an element of } U''; \text{ and if each of } x \text{ and } y \text{ is an element of } U', \text{ then } z = \rho'((x, y));$

or if x is an element of U' and y is not an element of U' , then $z = \{w : w \text{ is an element of } y, \text{ or there is an element of } y, \text{ call it } q, \text{ so that } w = \rho'((x, q))\}$;

or if x is not an element of U' and y is an element of U' , then $z = \{w : w \text{ is an element of } x, \text{ or there is an element of } x, \text{ call it } q, \text{ so that } w = \rho'((q,y))\}$ };

or if x is not an element of U' and y is not an element of U' , then $z = \{w : w \text{ is an element of } x; \text{ or there is an element of } x, \text{ call it } p, \text{ and there is an element of } y, \text{ call it } q, \text{ so that } w = \rho'((p,q))\}$ }.

Problem 23'': Show that ρ'' is a function from $\{(x,y) : x \text{ is an element of } U'' \text{ and } y \text{ is an element of } U''\}$ into U'' .

Definition 17'': $\tau'' = \{((x,y),z) : x \text{ is an element of } U'' \text{ and } y \text{ is an element of } U'', \text{ and if each of } x \text{ and } y \text{ is an element of } U', \text{ then } z = \tau'((x,y));$

or if x is an element of U' and y is not an element of U' , then $z = \{w : \text{there is an element of } y, \text{ call it } q, \text{ so that } w = \tau'((x,q))\}$ };

or if x is not an element of U' and y is an element of U' , then $z = \{w : \text{there is an element of } x, \text{ call it } q, \text{ so that } w = \tau'((q,y))\}$ };

or if x is not an element of U' and y is not an element of U' , then $z = \{w : \text{there is an element of } x, \text{ call it } p, \text{ and there is an element of } y, \text{ call it } q, \text{ so that } w = \tau'((p,q))\}$ }.

Problem 24'': Show that τ'' is a function from $\{(x,y) : x \text{ is an element of } U'' \text{ and } y \text{ is an element of } U''\}$ into U'' .

Problem 28'': Suppose that x and y are elements of U'' . Show that $\rho''((x,y)) = \rho''((y,x))$.

Problem 29'': Suppose that x and y are elements of U'' . Show that $\tau''((x,y)) = \tau''((y,x))$.

Problem 30'': Suppose that each of x, y , and z is an element of U'' . Show that $\rho''((\rho''((x,y)),z)) = \rho''((x,\rho''((y,z))))$.

Problem 31'': Suppose that each of x, y , and z is an element of U'' . Show that $\tau''((\tau''((x,y)),z)) = \tau''((x,\tau''((y,z))))$.

Problem 32'': Suppose that each of x, y , and z is an element of U'' . Show that $\tau''((x,\rho''((y,z)))) = \rho''((\tau''((x,y)),\tau''((y,z))))$.

Problem 35'': Suppose that x and y are elements of U'' , (x,y) is an element of G'' , and z is an element of U'' . Show that $(\rho''((x,z)),\rho''((y,z)))$ is an element of G'' .

Problem 42'': Suppose that x and y are elements of U'' so that (x,y) is an element of G'' . Show that there is exactly one element of U'' , call it q , so that $\rho''((x,q)) = y$.

Problem 43'': Suppose that x is an element of U'' . Show that there is exactly one element of U'' , call it y , so that $\tau''((x,y)) = (1,1)$.

Chapter 2

Examples

Following are some theorems that students proved during my Spring 1999 offering of the course, for which a selection from the problems above formed the corpus from which they worked. The problem from the notes which each addresses is noted in parentheses. The theorem stated may not solve that problem; the reference is intended to show what students might be thinking when working on the problem. The statements of the theorems typically include the structure the student created to address the referenced problem.

Theorem (6): Suppose that (a, b) and (c, d) are elements of U , and (a, b) precedes (c, d) . Then $((a * d) + (c * b), 2 * b * d)$ is an element of U so that (a, b) precedes $((a * d) + (c * b), 2 * b * d)$, and $((a * d) + (c * b), 2 * b * d)$ precedes (c, d) . (This is a nice example of using the arithmetic of the “midpoint” to find a candidate for the solution. It is also nice in that the element produced need not be in U' even when both (a, b) and (c, d) are, so there is still something to be discovered if a student were to try to use the construction that worked here on Problem 6'.)

Theorem (6'): Suppose that (a, b) and (c, d) are elements of U' so that (a, b) precedes (c, d) and so that $a < b$, $d < c$, (a, d) is an element of U' , and (c, b) is an element of U' . Then $(a * c, b * d)$ is an element of U' so that (a, b) precedes $(a * c, b * d)$ and $(a * c, b * d)$ precedes (c, d) . (Here is an example of a theorem that only partially solves Problem 6'. It is not unusual for students to case such problems before finding a solution such as the one above that handles all cases at once.)

Definition: Suppose that each of n and m is a natural number. k is the *greatest common factor* of n and m means that there is a natural number, call it p , so that $n = k * p$; and there is a natural number, call it q , such that $m = k * q$; and (p, q) is an element of U' . (The student here made a definition for a term with which he had pre-existing familiarity in terms of the structure of the course!)

Notation: $\gcd\{a, b\}$ stands for “the greatest common factor of a and b ”.

Theorem (25): Suppose that (a, b) is an element of U .

Then $\left(\frac{a}{\gcd\{a, b\}}, \frac{b}{\gcd\{a, b\}}\right)$ is an element of U' .

Theorem (30): Suppose that each of (a, b) , (c, d) , and (e, f) is an element of U' . Then

$$EC_{(a*((c*f)+(d*e)), b*(d*f))} = EC_{\left(a*\frac{(c*f)+(d*e)}{\gcd\{(c*f)+(d*e), d*f\}}, b*\frac{d*f}{\gcd\{(c*f)+(d*e), d*f\}}\right)}$$

Theorem (21): Define a by $a(1) = (3, 2)$, and if n is a natural number, then

$$a(n+1) = \left(\frac{\Pi_1 a(n) + 2*\Pi_2 a(n)}{\gcd\{\Pi_1 a(n) + 2*\Pi_2 a(n), \Pi_1 a(n) + \Pi_2 a(n)\}}, \frac{\Pi_1 a(n) + \Pi_2 a(n)}{\gcd\{\Pi_1 a(n) + 2*\Pi_2 a(n), \Pi_1 a(n) + \Pi_2 a(n)\}} \right)$$

Then a is a function from the natural numbers into U' .

Theorem (19): Suppose that $A = \{(p, q) : (p, q) \text{ is an element of } U' \text{ and } p*p < 2*q*q\}$ and $B = \{x : x \text{ is an element of } U' \text{ and } x \text{ is not an element of } A\}$. Then $B = \{(p, q) : (p, q) \text{ is an element of } U' \text{ and } p*p > 2*q*q\}$, and if (a, b) is an element of U' , then

$$\left(\frac{(3*a) + (4*b)}{\gcd\{(3*a) + (4*b), (2*a) + (3*b)\}}, \frac{(2*a) + (3*b)}{\gcd\{(3*a) + (4*b), (2*a) + (3*b)\}} \right)$$

is an element of U' so that if (a, b) is an element of A , then it is an element of A and (a, b) precedes it; or if (a, b) is an element of B , then it is an element of B and it precedes (a, b) .

Chapter 3

Remarks

The following comments contain information about my intent for many of the problems, and experiences that my students have had with them.

Definition 1: Notice that the idea of *domain* is included in the definition of *function*, by proviso (ii.).

Definition 2; Problem 1; Definition 4: I have written the words for definitions of the sets in question here. Students often translate these to the notation for these words suggested back on the page on sets. If they don't, I usually suggest that they see if they can.

Problems 1 and 3: The students I teach in this course are typically very naïve and seldom have been forced to deal with the need to say things carefully. I put Problems 1 and 3 in the notes to address the fact that early in every course, a student would define a “function” as $\{(x, y) : x \text{ is an element of } X \text{ and } y \text{ is an element of } Y\}$, and then claim whatever additional properties he/she needed, as he/she needed them. When it happens now, I can point to Theorem 1 and start the discussion there. Problem 3 reinforces “not every set of ordered pairs whose coordinates are in the right sets is a function”.

Problem 2: This problem is here to emphasize that in this set theory, sets have at least one element each. Discussion of the “empty set” will naturally occur here.

Definition 6: Notice that p and q represent ordered pairs here. This forces the students to take meaning from the words, and when arguing from or to Definition 6, gives an opportunity for them to rewrite their instantiations so that the coordinates are explicitly denoted.

Problems 4, 5, and 6: These are all properties assumed about $<$ on $\mathfrak{R}$. Although U will be modified when it fails to support precedes as an order, the proofs made in U for 4, 5, and 6, properly adjusted, usually go over to all subsequent modifications.

Problem 7: This is the first false conjecture, since, for instance, $(2, 3)$ and $(4, 6)$ cannot be compared by *precedes*. Historically, the resolution to this problem was to identify all such objects as representing the same thing, an idea formalized with equivalence

classes. I prefer to emphasize that in a model, different objects must be different, and to continue with a subset of the objects with which we are working. (In “real life”, i.e. the world of computation, these are the only objects which are accessible. Equivalence classes come up again and again anyway, so the idea is still there to be studied later.)

Definition 7: *Commands* is the concept central to counting. The Schröder-Bernstein Theorem, which is not addressed in this course, guarantees that it is sufficient to admit the classical results.

Problems 9 and 10: Although all students in your class will “know” that *tangent* maps $(-\frac{\pi}{2}, \frac{\pi}{2})$ onto $\mathbb{R}$, they will likely not realize that this precludes Problem 10 from being a theorem. Indeed, most of the time, classes try to prove Problem 10 and the argument typically includes as a punch line something equivalent to “because the containing set contains more elements than the subset”. This affords a marvelous opportunity to teach the difference between ordinary language and formal language, since “more” in the subset sense turns out to be different than “more” in the counting sense (sometimes!). The questions that show incorrect arguments are incorrect often lead to the example $n + 1 \rightarrow n$.

Problem 11: If Problem 7 has been done at this time, use Problem 11’ here instead.

Problem 12: This problem virtually guarantees a case argument will be forthcoming.

Problem 13: Since Problem 7 is not a theorem, if this problem is solved, then it will be done using the order structure. Thus, together with Problem 12, we get that segments defined by an order with neither max nor min create a basis for a Hausdorff topology on the set on which the order is defined.

Definitions 13, 14, and 15: In an axiom system for the numbers, one has (at least) the choice of the greatest lower bound property, the Bolzano-Weierstrass property, the Heine-Borel property, or the Dedekind cut property, as a completeness axiom. I choose the Dedekind cut property, since it can be articulated without reference to any structure other than the order itself. This is the first idea in the course that the students are likely not to have encountered in another context.

Problem 14: An idea for proving Problem 14 has usually already been addressed by this point in the course. Usually students will argue as if they have “ (a, b) or (b, a) , and not both” instead of “ (a, b) or (b, a) ” in the definition of order, and can be forced to address whether or not this is a consequence of the definition at that time.

Problems 15 and 16: These problems are usually solved using cuts exhibiting the Dedekind cut property, thus establishing a pretext for asking “must all cuts be like these?”

Problem 17: As we find it in counting, the order “ $<$ ” does not have the Dedekind cut property.

Definitions 4’ and 6’, and Problems 4’ - 8’: Problem 7 will be solved at some point. The example that shows Problem 7 is not a theorem will show two elements, neither of which precedes the other. Often students demonstrate that properties (i.) and (iii.) from Definition 3 hold before providing the example that shows that property (ii.) fails. Sometimes a single example, such as $(1, 1)$ and $(2, 2)$, is offered; sometimes

students describe the phenomenon that is in Definition 4'. The instructor needs to make the structure in Definition 4' plausible from whatever platform the students provide. This may necessitate stating more problems, or initiating a class discussion in which other examples are created, and the structure identified in Definition 4' is shown to be common to all of them.

I have placed these problems after Problem 17 *only* because it has been typical in my experience that Problem 7 gets solved before Problems 15 - 17 get solved. Whenever Problem 7 gets solved, it is time for Definitions 4' and 6', and Problems 4' - 8'. Until Problem 7 gets solved, it is not time for Definitions 4' and 6', or Problems 4' - 8'. If students have discovered the "reduction argument" within equivalence classes, all five (six) problems will be direct consequences of it using the arguments from 4 - 8 (and 11) to get the objects from which to reduce. If students have not discovered the "reduction argument", an interesting sidelight is to see which constructions from 4 - 8 give elements of U' . These problems offer an outstanding context to point out the power of "some arguments", and always afford at least an opportunity to show how to modify an argument to meet new conditions.

Problems 18 and 19: Here is an example which shows that G' fails to have the Dedekind cut property.

Problems 20 - 22: Once Problem 10 gets solved, and the possibility of seemingly "smaller" sets commanding seemingly "larger" sets is established, these problems are ready to be addressed. If U' has not been defined yet, state Problems 21 and 22 using U instead of U' , and save " U' commands U " and the appropriate modifications of 21 and 22 for when U' is defined. Also, the problem can be stated with its correct conclusion to direct students, or with the incorrect conclusion to misdirect students; I choose this particular presentation in order to illustrate that sometimes there is nothing in previous experience to indicate what the appropriate conjecture is.

Definitions 16 and 17; Problems 23 and 24: These are the algorithms for adding and multiplying fractions, but neither problem is a theorem. Both ρ and τ , however, are functions from $U' \times U'$ into U . In my experience, whenever I state algebra problems, the majority of the class becomes enamored of them and work on little else without severe prodding. I typically postpone stating them as late as I can. These are nice in that the counterexamples can be used to suggest that even though the "answers" are different, they are "related" to the same element of U' , and thus yet another opportunity is afforded for discovering the pertinent equivalence classes.

Definition 18, Problems 25 - 27: This definition and these problems are appropriate whenever the idea becomes clarified. Sometimes it happens when Problem 7 is settled; usually it happens by the time the U and G theorems are reproven for U' and G' . If it hasn't happened at this stage, they need to be stated since they are important for the algebra problems.

Problems 23', 24', and 28 - 34: These problems investigate the status of the field axioms. Problems 23' and 24' are closure properties, 28 and 29 are commutative properties, 30 and 31 are associative properties, 32 is the distributive property, and 33 and 34 are identity properties. Of these, 33 is the only one which is not a theorem; indeed, if 35 were to be proven before 33, it shows that 33 cannot be true.

Problem 35: This is the property that addition preserves order.

Problems 36 and 37: Every equation for addition that can have a solution does have one, and the reciprocal property holds for multiplication. 36 is appropriate once 33 is shown not to be a theorem and 35 is proven. 37 is appropriate once 34 is proven. If a class were to get this far, it would have established a model for the positive rational numbers and its algebra.

Definition 4'': Be aware that Definition 4'' introduces a new type of element to the model. Each element of U' is still there "... x is an element of U' , or ...", but there are also elements which are subsets of U' "there is a cut of U' by G' , call it (A,B) , so that A has no maximum by G' and B has no minimum by G' , and $x = A$ ". Dedekind made each number one of these (all you need do is drop the " A has no maximum and B has no minimum"); I prefer the spirit of computation. These numbers are there, but our computational access to them is through our access to U' , a denumerable set dense in U'' by G'' . This is analogous to the situation in the decimal model where even though infinite decimal expansions exist, our computational access to them is typically through the arithmetic of numbers with terminating decimal expansions.

Definition 6'': G'' on U'' models " $<$ " on the numbers.

Problems 4'', 5'', 6'', 7'', 21, and 38: These problems demonstrate that G'' on U'' models " $<$ " on the numbers. Note that in 6'', the problem asks for an element of U' , which together with 21, gives a countable dense set. Geometrically, the structure may as well be the numbers. The lack of an additive identity for the algebra indicates that the model is more suitable for " $<$ " on the positive numbers.

Problems 39 - 43: Since any order with the Dedekind cut property has the "betweenness" property from which 40 is typically proven, these problems give a template for a proof that any order with the Dedekind cut property must be uncountable.

Problems 44 - 51: Prepare to extend the addition and multiplication to U'' .

Problems 23'' and 24'': The structure for extending the algebra is done in 44 - 51, but the student still must prove that the complement of " C " has no min.

Problems 23'', 24'', 28'' - 32'', 35'', 37'' and 38'': These establish that τ'' and ρ'' have the structure that is assumed for $*$ and $+$ on the positive numbers. If one desires to complete U'' , G'' , τ'' , and ρ'' to a model for the numbers, $<$, $*$, and $+$; all that is needed is to introduce an object to be 0, and for each element of U'' , an object that can be identified with it (I typically reserve the symbol $\tilde{x}$ for the object to be identified with x).

The order is extended by letting 0 precede every element of U'' , every element of $\tilde{U}''$ precede 0, and letting elements of $\tilde{U}''$ reverse the order of the elements from U'' from which they are built. τ'' is extended by using the results on U'' and supplying a rule of "signs"; $\tau''((x,y))$ is in $\tilde{U}''$ means exactly one or x and y is in $\tilde{U}''$ and the object for the answer comes from looking at τ'' as applied to the objects in U'' from which x and/or y are built.

Problems associated with these ideas are likely to be routine for a class that has gotten this far. I tend to use problems associated with these ideas as exam questions.

Extending ρ'' is more interesting, since it involves using solutions to equations on U'' (those addressed in 42'') for defining sums.

Chapter 4

Some Axioms for the Numbers

The primitive words are “number”, “ $<$ ”, “ $+$ ”, and “ $*$ ”.

Axiom G1: “ $<$ ” is an order on the set of numbers.

Axiom G2: It is not the case that the set of numbers has a min by “ $<$ ”, and it is not the case that the set of numbers has a max by “ $<$ ”.

Axiom G3: There is a sequence in the set of numbers, call it Q , such that if x and y are numbers, then there is a natural number, call it k , so that $x < Q(k)$ and $Q(k) < y$.

Axiom G4: “ $<$ ” has the Dedekind cut property.

Axiom A1: If each of x and y is a number, then $x + y$ is exactly one number, and $x * y$ is exactly one number.

Axiom A2: If each of x and y is a number, then $x + y = y + x$, and $x * y = y * x$.

Axiom A3: If each of x , y , and z is a number, then $x + (y + z) = (x + y) + z$, and $x * (y * z) = (x * y) * z$.

Axiom A4: 0 is a number so that if x is a number, then $0 + x = x$; and 1 is a number so that if x is a number, then $1 * x = x$.

Axiom A5: If x is a number, then there is exactly one number, call it y , so that $x + y = 0$; and if x is a number different than 0, then there is exactly one number, call it w , so that $x * w = 1$.

Axiom A6: If each of x , y , and z is a number, then $x * (y + z) = (x * y) + (x * z)$.

The Combining Axiom: If x and y are numbers so that $x < y$, and w is a number, then $x + w < y + w$.